

Prove your C code free of run-time errors.

For every input, every path and every thread interleaving — or get the exact input that makes it fail. Automatically, with a certificate your auditors can check.



Every input, not a sample

Testing checks the inputs you thought of. The verifier proves the absence of errors for all of them — 2^{32} values of an int, every loop iteration, every schedule of your threads.



Real bugs, with the failing input

When a property does not hold, you get a concrete input and a step-by-step execution, replayed on the exact semantics of C — never a false alarm.



Evidence, not opinions

A PDF certificate with the proof drawn on your program, plus JSON, SARIF and the complete proof for independent re-checking — ready for reviews and safety cases.

What it proves

Property	CWE
Assertions and functional properties you state	CWE-617
Signed integer overflow	CWE-190
Division by zero	CWE-369
Invalid shifts	CWE-1335
Array index out of bounds	CWE-787
Null, dangling or out-of-bounds pointers	CWE-476, 416
Double or invalid free	CWE-415, 761
Memory leaks	CWE-401

Programs

Sequential C, recursive functions, multithreaded code (POSIX threads, mutexes, C11 atomics, lock-free protocols), multi-file projects, single functions verified for every caller.

Where it matters

Automotive (ISO 26262), avionics (DO-178C), industrial control (IEC 61508), medical devices, protocol parsers, security boundaries, and libraries with many callers.

Fits your workflow

One command line, exit codes for CI gates, SARIF for GitHub and GitLab code scanning, and a C++ SDK to embed it in your own tools.

Proven on 177 benchmarks — no wrong answer

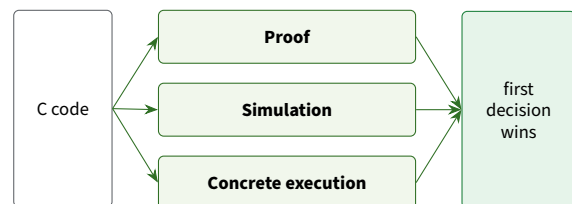
The trial package contains every one of these benchmarks and the verifier’s result for it (certificate or counterexample report, JSON, SARIF, proof). Each is a correct result against the known answer: **102** programs proven free of the checked errors, **75** real bugs found with the failing input.

Category	Benchmarks	Proven safe	Bug found	Median time
Sequential (SV-COMP)	47	33	14	10 s
Recursive (SV-COMP)	92	48	44	10 s
Concurrent (SV-COMP and thread suite)	24	12	12	12 s
Tutorial examples	14	9	5	8 s
Total	177	102	75	

Median time per benchmark includes writing the PDF (about 8 s); 4-core laptop, 60-second budget.

How it works

A portfolio of techniques races on every check, in parallel processes: an automatic **proof engine** (invariants, function contracts, Owicki–Gries reasoning for threads, counterexample-guided refinement), a **random interleaving** of threads, and a byte-exact **concrete execution**. The first to decide a check settles it. Proofs are *sound* — the model over-approximates the program — and every counterexample is *validated* against the exact semantics before it is reported.



Editions

	Trial	Professional license
Verification engine	complete	complete
Runs	1000 within 50 days	as licensed — up to unlimited for the period
Certificates and reports	marked “Trial version”	your license number and company name on every document
PDF protection	—	each PDF protected by its own password
License	self-service download	bound to your company: name, address and password
Support	e-mail	priority support; verification services on request

Try it today, license it when you are convinced.

The trial package includes everything on these pages. For licenses, volume pricing, and verification services by Giolit’s engineers (proofs of your critical modules, certificates for safety standards, training):

 www.giolit.com  business.giolitlabs@gmail.com

Giolit Labs, Ramavarmapuram P.O., Thrissur, Kerala, India

